

Explainable Automatic Modulation Recognition using CNN–GRU–GNN with Curriculum Learning and Retrieval-Augmented Generation

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Abstract—Automatic Modulation Recognition (AMR) is a vital component of the intelligent wireless communication systems that allow the adaptive transmission, monitoring of the spectrum, and cognitive radio applications. Deep learning based classification models have progressed the accuracy of classification, but most of the current techniques have poor robustness under low Signal-to-Noise Ratio (SNR) and are difficult to interpret.

In this paper, we suggest an explainable AMR framework incorporating a hybrid CNN–GRU–GNN model, curriculum learning, adaptive modulation strategy, and retrieval augmented generation (RAG). CNN learns spatial features from raw I/Q signals, GRU learns their temporal dependencies, and GNN learns their structural relationships in signal sequences. The two-stage curriculum learning strategy is used, with the model first trained with high SNR samples and then fine-tuned with all the data to make it more robust in the presence of noise.

Gradient-based saliency maps and a hybrid RAG model employing FAISS with BM25 retrieval are adopted to provide communication-theoretic explanations for model predictions, facilitating interpretability. Results from the RadioML 2016.10A dataset illustrate the robustness and accuracy of the proposed framework and confirm that the proposed framework outperforms the current state of the art approach in terms of classification accuracy for high SNR conditions, while maintaining better robustness for SNR < 0 dB conditions.

The proposed system can effectively achieve the balance of accuracy, robustness and explanation of the next generation adaptive wireless communication system.

Index Terms—AMR, CNN-GRU-GNN, Explainable AI, RAG, Wireless Communication, SNR

I. INTRODUCTION

Wireless communications systems require an important task, namely Automatic Modulation Recognition (AMR), to perform adaptive transmission, spectrum monitoring and cognitive radio applications. Without the knowledge of the transmitted modulation scheme, it is impossible to identify the modulation scheme.

The traditional AMR methods are based on handcrafted feature extraction and likelihood-based methods, which fail to work efficiently in the case of noisy channels. AMR has been greatly improved with the use of deep learning, where an end-to-end approach to feature learning is obtained directly from the raw in-phase and quadrature (I/Q) signals [2]. The ability of CNN to learn spatial features is beneficial [3] and the ability of CNN-LSTM to model temporal features is beneficial [3], [7], [8].

In recent years, SNR-aware learning [5], multitask learning [6], [24] and transformer-based learning [22] have been studied to address the robustness under different channel conditions. But many current models still exhibit poor performances at low SNRs and lack interpretability.

In the field of wireless signal classification, explainable AI techniques have been applied [11]–[13] and Retrieval-Augmented Generation (RAG) frameworks [14]–[17] have been developed for providing knowledge-grounded reasoning capability. The use of AMR systems in recent AI-driven communication systems [21], [23], [25] underscores the demand for effective and comprehensible AMR systems.

To tackle these problems, this paper suggests an explainable AMR framework that combines hybrid CNN–GRU–GNN architecture, curriculum learning, adaptive modulation strategy and explainability based on RAG. It adopts CNN to extract spatial features, GRU to model temporal information, and GNN to learn the relationships between structures. The learning strategy adopted is a two-stage curriculum learning process by training the model using high SNR samples and fine-tuning the model using the full data set to get more adversarial learning.

The proposed framework is a mixture of deep learning and knowledge-grounded reasoning, as shown in Fig. 3, in order

to achieve accurate and interpretable modulation recognition.

The major contributions of this work are as follows:

- A hybrid CNN–GRU–GNN architecture for AMR systems
- A curriculum learning strategy for improved low-SNR robustness
- A hybrid RAG-based explainability framework using FAISS and BM25 retrieval
- An adaptive modulation strategy for intelligent wireless communication

II. LITERATURE REVIEW

The RadioML 2016.10A dataset [1] is adopted as a standard dataset for testing Automatic Modulation Recognition (AMR) systems and is characterized by a wide range of Signal-to-Noise Ratio (SNR) and a wide variety of modulation schemes. O’Shea and Hoydis [2] demonstrated that deep learning models can learn discriminative features directly from the I/Q data instead of using conventional handcrafted methods.

AMRs are commonly used using Convolutional Neural Networks (CNNs) due to their ability to extract spatial signal features [3]. Hybrid CNN-LSTM networks [3], [7], [8] were also developed to take advantage of the temporal dependency and Han *et al.* [4] proposed work that increased the recognition by incorporating the fusion of signal representation and constellation diagrams.

There have been recent studies that investigate SNR-aware learning [5], multitask learning [6], [24] and transformer-based methods [22] for enhancing robustness under different channel conditions. But the models of transformers can be complicated. Robustness, efficiency and interpretability are major challenges identified in the survey studies [9], [10] on deep learning based AMR systems. The adaptive and intelligence aspects of AMRs are also emphasized in the context of communication research using AI technology [21], [23], [25].

Explainable AI is now an emerging field of research for wireless signal classification. Xu *et al.* [11], Elgendy *et al.* [12] and Singh *et al.* [13] investigated the interpretable deep learning approaches for communication systems. Most of the current approaches, however, are based in the saliency or attribution approach without communication-theoretic knowledge.

Another important challenge is robustness with respect to noisy and adversarial environments. Sadeghi and Larsson [18] showed vulnerabilities of the deep learning-based AMR systems against adversarial attacks and later studies [19], [20] focused on robust training techniques to ensure reliable signal classification.

Knowledge Grounded Reasoning tasks have seen retrieval-augmented generation (RAG) frameworks perform well, such as [14]–[17]. But their use with AMR systems is still largely unknown.

A. Research Gap

Currently, most AMR approaches are limited to the optimization of either accuracy or robustness or explainability.

CNN-based methods are able to extract spatial features well, while they do not model temporal information, and CNN-LSTM models do not explicitly model the relationships among signal components. Transformer-based methods boost representation learning and incur high computational costs.

Further, the current explainable AI methods are primarily focused on saliency-based interpretation and lack communication-theoretic knowledge. Most AMR systems are also unable to adapt the modulation when the channel conditions are changing.

These limitations inspire the proposed framework that combines CNN, GRU, and GNN for multi-level feature learning, curriculum learning to boost robustness, adaptive modulation approach, and hybrid RAG-based explainability.

III. DATASET AND PREPROCESSING

The proposed framework is tested with the RadioML 2016.10A dataset [1] which is wellknown as a benchmark for Automatic Modulation Recognition (AMR). The wireless signal samples in the dataset are generated synthetically under realistic channel conditions with Signal-to-Noise Ratio (SNR) ranging from -20 dB to 18 dB and are about 220,000 in number.

The dataset includes multiple analog and digital modulation schemes such as:

- | | | |
|---------|---------|----------|
| • BPSK | • QAM64 | • AM-DSB |
| • QPSK | • PAM4 | • AM-SSB |
| • 8PSK | • CPFSK | • WBFM |
| • QAM16 | • GFSK | |

Each sample is represented as a complex in-phase and quadrature (I/Q) signal with shape (2,128), where the two channels correspond to the in-phase and quadrature components respectively.

A. Exploratory Data Analysis

Exploratory analysis of the dataset was done in order to understand the behavior of the signal for different channel conditions. Figure 1 shows the constellation representations for various SNRs. The constellation points are strongly influenced by noise at lower SNR, and less at higher SNR.

Fig. 2 shows a sample I/Q signal representation from the dataset, highlighting temporal variation in wireless signal components.

The exploratory analysis shows the separation of signals is greatly enhanced by SNR increase, which justifies using the curriculum learning for robust AMR.

B. Preprocessing

All signal samples are normalized into the [-1,1] range before training, to reduce the impact of using large gradients on the optimization process and help it converge. Supervised learning with the labels represented in the dataset as numerical class representations.

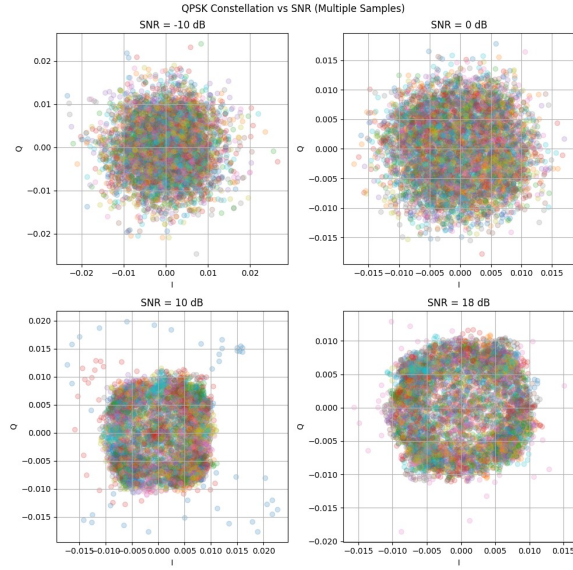


Fig. 1. Constellation Diagram under Different SNR Levels

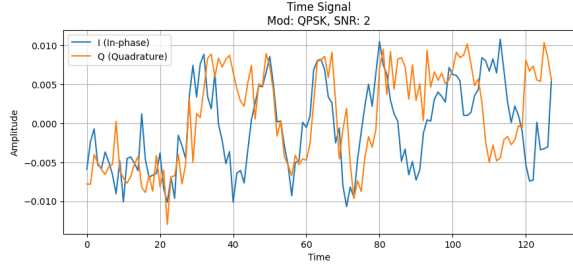


Fig. 2. Sample I/Q Signal Representation

Stratified sampling is performed to ensure that the dataset is balanced across classes, modulation schemes, and SNR. The training, validation and testing ratios are 70:15:15 finally.

Curriculum learning is used for learning during training to enhance robustness. First training is achieved by using samples that are more likely to contain a "cleaner" constellation pattern at high SNRs. Next, fine-tuning is done with the entire data set, including low-SNR samples, allowing the model to adjust to noisy communication channels.

IV. PROPOSED METHODOLOGY

The proposed Explainable Automatic Modulation Recognition (AMR) framework combines deep learning, curriculum learning, adaptive modulation, and explainability grounded on knowledge to robustly and interpretable classify wireless signals.

The architecture includes a hybrid CNN-GRU-GNN, as shown in Fig. 3, and a hybrid Retrieval-Augmented Generation (RAG) module. This full pipeline involves signal preprocessing, multi-level feature extraction, classification of modulation, adaptive modulation recommendation and generation of explainability.

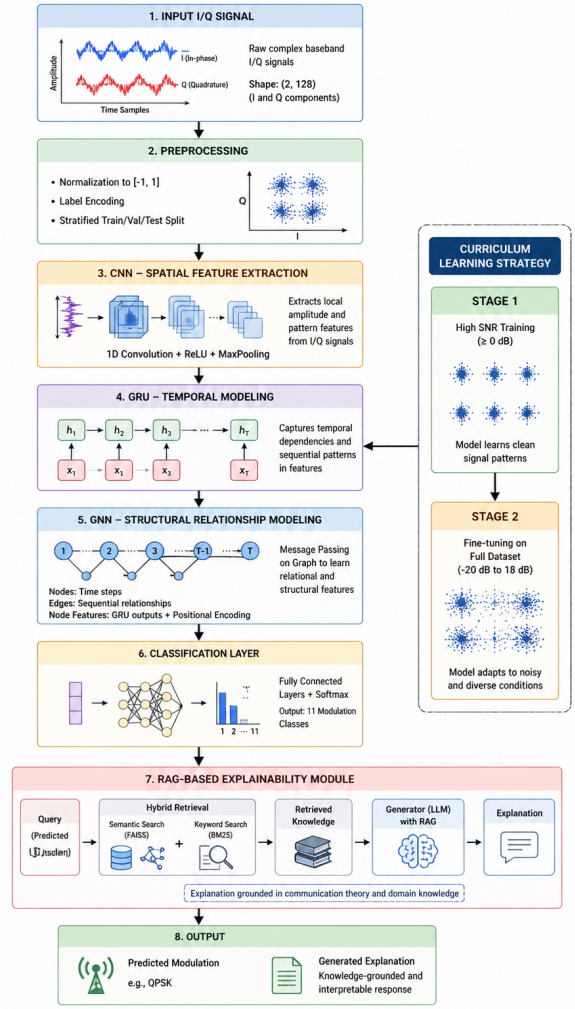


Fig. 3. Proposed CNN-GRU-GNN with Hybrid RAG Framework

A. Signal Representation

The complex in-phase and quadrature components are used to represent each wireless signal sample:

$$x(t) = I(t) + jQ(t) \quad (1)$$

where $I(t)$ and $Q(t)$ are the in-phase and quadrature components, respectively. Samples of the data set are:

$$X \in \mathbb{R}^{N \times 2 \times 128} \quad (2)$$

where N represents the total number of signal samples.

B. CNN-Based Spatial Feature Extraction

The raw I/Q signals are fed into the one-dimensional Convolutional Neural Network (CNN) to obtain local spatial characteristics. The convolution operation is given by:

$$F = \sigma(W * X + b) \quad (3)$$

W is a set of convolution filters and b is a set of bias parameters while the ReLU activation function is denoted

by σ . The CNN receives spatial patterns of local amplitude changes and constellation.

C. GRU-Based Temporal Modeling

The CNN features extracted are fed into a Gated Recurrent Unit (GRU) network for modeling temporal dependencies in the wireless signal sequence:

$$h_t = GRU(F_t, h_{t-1}) \quad (4)$$

where h_t is the hidden state at time step t . GRU has a lower computational complexity than LSTM-based methods [3], [7] and is able to retain sequential information.

D. Graph Neural Network (GNN)

The outputs of the GRU are fed into the graph as graph nodes to represent structural relationships between signal components. Time steps are connected in a sequence by graph edges.

The message passing on the GNN is done as follows:

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} W h_j^{(l)} \right) \quad (5)$$

where neighbors of node i are denoted by $\mathcal{N}(i)$. This allows the model to learn the relationship between signal sequences.

E. Curriculum Learning Strategy

A two-stage curriculum learning strategy is used for enhancing channel robustness under channel variations:

- Stage-1: Training using high-SNR samples (≥ 0 dB)
- Stage-2: Fine-tuning using the complete dataset including low-SNR samples

The first stage allows the model to learn more accurate 'cleaner' constellation patterns and to attain about 93% classification accuracy. The second stage provides robustness in noisy communication and also boosts the performance for SNR_i0 dB.

In this section, the classification and adaptive modulation are explained. In this section, the classification and adaptive modulation are described.

The last learned representation is fed into a softmax classifier:

$$\hat{y} = \text{Softmax}(Wh + b) \quad (6)$$

where $\hat{y}$ is predicted modulation classes.

The framework can also suggest adaptive modulation schemes based on the estimated SNR conditions:

If the signal-to-noise ratio is low, use BPSK instead. If the SNR is low, then use BPSK instead.

- Low SNR $\rightarrow$ QPSK / 8PSK When SNR is high, the constellation is QAM. If the SNR is high, the constellation is QAM.

This allows an adaptive communication functionality by using intelligence in dynamic wireless environments [23], [25].

Saliency maps can be used to make AI decisions understandable. AI decisions can be explained using saliency maps.

Gradient-based saliency maps are employed to highlight regions of the input signal that drive model predictions, thereby improving the interpretability of the model.

$$S = \left| \frac{\partial \hat{y}}{\partial X} \right| \quad (7)$$

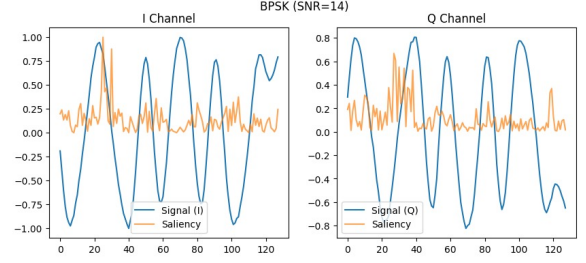


Fig. 4. Saliency Map Fine-Tuned Model

The model concentrates on relevant signal regions that are important for modulation classification as illustrated in Fig. 4.

F. Retrieval-Augmented Generation (RAG) Framework

The proposed system combines a hybrid Retrieval-Augmented Generation (RAG) approach to assist in the generation of communication-theoretic explanations in addition to modulation classification. Information pertaining to modulation theory, I/Q signal representation, SNR behavior, constellation characteristics, spectral efficiency, adaptive modulation strategies and channel impairments were used to build a domain-specific knowledge base.

The framework employs two types of retrieval: dense semantic retrieval with FAISS vector indexing and transformer-based embeddings, as well as sparse keyword-based retrieval with BM25 algorithm to improve retrieval quality. For both of the above retrieval methods, outputs are fused to enhance contextual relevance, using Reciprocal Rank Fusion (RRF).

Inference, the predicted modulation class and the estimated SNR condition are used as retrieval queries. This retrieved contextual information is then provided to a Large Language Model (LLM) for the generation of human-understandable explanations for modulation characteristics and adaptive communication behavior.

$$z = \text{Retriever}(q); y_{exp} = \text{Generator}(z, \hat{y}) \quad (8)$$

In which z stands for retrieved communication knowledge and y_{exp} stands for generated explanation. The proposed RAG framework significantly extends the interpretability by building on the communication theory, not just based on the black box neural predictions [14]–[17].

V. RESULTS AND ANALYSIS

The proposed scheme is tested on RadioML 2016.10A data set [1] with different Signal to Noise Ratios (SNR). Classification accuracy, confusion matrix evaluation and SNR-wise robustness analysis are used to analyze the performance.

A. Model Performance

There are two different models configurations that are assessed:

The model trained under high SNR is called a model-1. Fine Tuned Model trained on the full dataset (Model-2)

The high-SNR trained model provides the best classification accuracy of about 93%, while the fine-tuned model shows better noise channel robustness.

TABLE I
MODEL PERFORMANCE COMPARISON

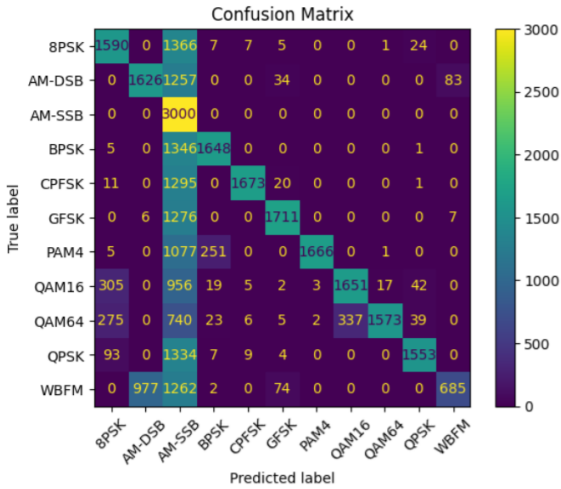
Model	Accuracy	Description
CNN	53%	Spatial feature extraction
CNN-LSTM	56%	Temporal modeling
CNN-GRU-GNN	93%	Proposed model

Table I shows that the proposed CNN-GRU-GNN model performs significantly better than the conventional CNN and CNN-LSTM model, which combines spatial, temporal, and structural feature learning. Similar improvements have been reported in recent AMR studies [6], [24] as well.

The proposed framework provides explainability and adaptive modulation capability that most existing frameworks lack, and, as demonstrated in Table II obtains better classification performance.

B. Confusion Matrix Analysis

The confusion matrix of the high-SNR trained model is given in Fig. V-B. The model can achieve a better classification accuracy for modulation schemes with better constellation separation.



The confusion matrix for the trained model at high SNR is shown below. The confusion matrix for trained model at high SNR is

The confusion matrix fine-tuned on the whole dataset is shown in Fig. V-B. The tuned down model shows better robustness in low SNR and less inter-class misclassification.

The robustness boost gained by fine-tuning corroborates previous work with curriculum learning and robust AMR systems [5], [20].

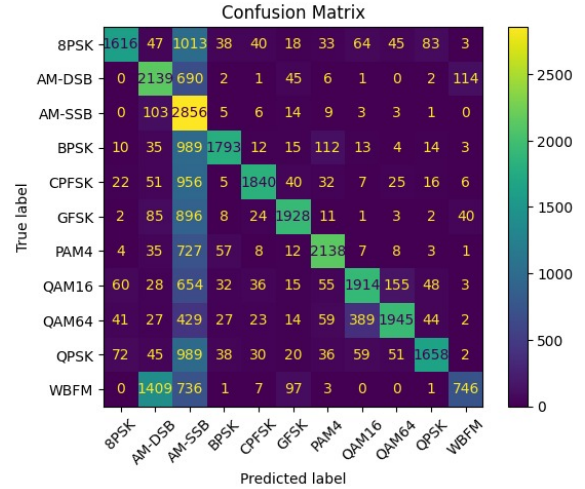


Figure 7

7 shows the confusion matrix for Fine-Tuned model. The confusion matrix of Fine-Tuned model is presented in Figure 7.

C. SNR-wise Performance Analysis

The classification accuracy is examined for both the model configurations for its robustness under different channel conditions by performing SNR-wise classification.

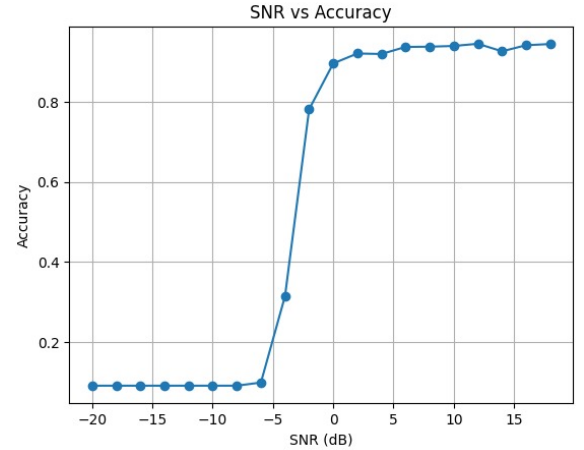


Fig. 5. SNR vs Accuracy for High-SNR Trained Model

The performance of the high-SNR trained model is very good in cleaner channel conditions while its accuracy is less at lower SNR, as shown in Fig. 5. Fine-tuning with the entire dataset, however, is shown to significantly boost robustness in the presence of noise in the case of Fig. 6.

The behaviour observed is consistent with the communication-theoretic principles and previous SNR-aware AMR studies [5], [6]. A low SNR tends to contribute to signal distortion and noise to the structure, while a higher SNR has better signal separation and classification accuracy.

D. Explainability Analysis

Saliency map in Fig. 4 delineates the important signal regions that are used for modulation classification decisions.

TABLE II
COMPARISON WITH EXISTING AMR MODELS

Ref.	Model	Key Idea	Accuracy
[3]	CNN-LSTM	Spatial + temporal learning	~60%
[5]	SNR-aware learning	Noise-adaptive classification	~70%
[6]	Multi-task CNN-LSTM	Joint optimization	~75%
[7]	Deep CNN-LSTM	Sequential feature learning	~80%
[22]	Transformer-based AMC	Attention learning	~85%
This Work	CNN-GRU-GNN + RAG	Explainable multi-level learning	93%

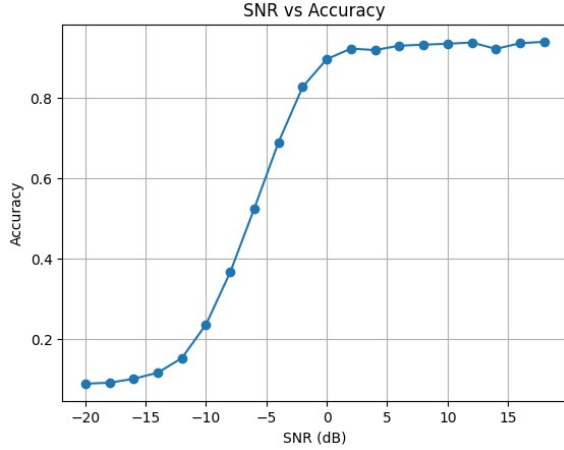


Fig. 6. SNR vs Accuracy for Fine-Tuned Model

The model emphasizes discriminative temporal and structural signal components to enhance interpretability of the deep learning framework.

VI. DISCUSSION

The proposed CNN-GRU-GNN framework combines spatial, temporal, and structural feature learning, thereby improving the performance of modulation recognition. Table I and Table II demonstrate that the proposed model performs better than the traditional CNN and CNN-LSTM models.

The curriculum learning strategy enhances the robustness in noisy channel. The fine-tuning on the full dataset improves the low SNR performance as seen in Fig. 5 and Fig. 6. Likewise, the confusion matrices in Fig. V-B and Fig. V-B indicate minimised misclassification among the various classes after fine-tuning.

The saliency map in Figure 4 and the hybrid RAG framework account for model predictions in terms of communication theory, enhancing the interpretability of the model [11]–[13].

VII. CONCLUSION

In this paper, we proposed a CNN-GRU-GNN (GNN) architecture-based explainable Automatic Modulation Recognition framework, curriculum learning, adaptive modulation strategy, and hybrid explainability using RAG framework.

Fine-tuning and experimental evaluation on the RadioML 2016.10A database showed that the algorithm achieves an accuracy rate of around 93% at high SNRs and achieves improved robustness at low SNRs. The suggested framework is a good balance between accuracy, robustness and interpretability in intelligent wireless communication.

Future work will focus on implementing SDR in real-time, deployment to edge devices and enhancement of adversarial robustness.

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